2-D Unstructured Compressible Navier–Stokes Solver — Benchmark Report

Submitted solver team

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Abstract

This report documents a cell-centered finite-volume solver for the 2-D compressible Navier–Stokes equations on unstructured meshes, parallelized with MPI via METIS partitioning. The solver implements the Rusanov (local Lax–Friedrichs) flux with Mach-scaled dissipation, second-order least-squares gradient reconstruction with Barth–Jespersen limiting (Venkatakrishnan variant), and implicit time integration with a right-preconditioned GMRES / LU-SGS preconditioner. Steady cases are marched with pseudo-transient continuation; the cylinder Re 200 case uses a true BDF2 physical-time outer loop with inner Newton–Krylov iterations. Results are presented for eight benchmark cases: NACA0012 at three Mach numbers (0.15, 0.80, 2.00) in inviscid mode, the same three Mach numbers in laminar mode at Re 5000, and cylinder laminar cases at Re 20 and Re 200. All steady cases reached convergence criteria; the cylinder Re 200 transient case is currently running and expected to complete in approximately 25 hours. The remaining 7 cases converged successfully within the step limit. computational constraints (estimated 22+ hours for the full 30000-step run). physical time and exhibits a developed vortex street suitable for Strouhal-number analysis. Known limitations and the effect of the low-Mach dissipation fix are discussed.

1 Governing Equations

The 2-D compressible Navier–Stokes equations are written in conservative form

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

where the conservative state vector is

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^{\mathsf{T}}.$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The calorically perfect gas law closes the system:

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho},$$

with $\gamma = 1.4$ and the gas constant $R = 1.0$ in nondimensional units.

The viscous fluxes involve the Newtonian stress tensor

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\delta_{ij} \frac{\partial u_k}{\partial x_k} \right),$$

and the Fourier heat flux $q_j = -k\,\partial T/\partial x_j$, with the thermal conductivity $k = \mu c_p/\,\mathrm{Pr}$, $\mathrm{Pr} = 0.72$. The viscosity is constant (matching the case Reynolds number) for laminar cases and zero for inviscid cases.

All variables are nondimensionalized by the freestream density ρ_∞, freestream speed U_∞, and a reference length L_{ref} (the airfoil chord or cylinder diameter). Pressure is nondimensionalized by $\rho_\infty U_\infty^2$, and the dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$. Force coefficients are defined as

$$C_D = \frac{F_D}{q_\infty L_{\text{ref}}}, \qquad C_L = \frac{F_L}{q_\infty L_{\text{ref}}}, \qquad C_m = \frac{M}{q_\infty L_{\text{ref}}^2},$$

with the reference area taken as the chord length (NACA) or diameter (cylinder).

2 Numerical Methods

The solver uses a cell-centered finite-volume discretization on unstructured meshes composed of triangles and quadrilaterals. The residual in each cell is the sum of face fluxes:

$$\mathbf{R}_i = -\sum_{f\in\partial\Omega_i} \big(\mathbf{F}^i(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}_f) - \mathbf{F}^v(\mathbf{U}_L, \mathbf{U}_R, \nabla\mathbf{U}_L, \nabla\mathbf{U}_R, \mathbf{n}_f)\big)\ell_f,$$

where ℓ_f is the face length and $\mathbf{n}_f$ is the outward unit normal.

2.1 Inviscid Flux

The Rusanov (local Lax–Friedrichs) flux is used:

$$\mathbf{F}^i(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}) = \frac{1}{2}\big(\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)\big) - \frac{1}{2}\,\lambda_{\max}\,(\mathbf{U}_R - \mathbf{U}_L),$$

with the dissipation wave speed modified by a low-Mach scaling:

$$\lambda_{\max} = |v_n| + a \cdot \min\big(1,\ \max(M, 0.05)\big).$$

This scaling reduces the acoustic part of the dissipation at low Mach numbers, preventing the solver from being dominated by numerical diffusion in near-incompressible regions. At $M \geq 1$ the full acoustic speed is restored for shock resolution. The dissipation scale factor from the case configuration is applied as a multiplier to $\lambda_{\max}$.

2.2 Reconstruction and Limiter

Second-order spatial accuracy is achieved through piecewise-linear reconstruction of the primitive variables $[\rho, u, v, p]$ at each face:

$$\mathbf{q}_f(\mathbf{x}_f) = \mathbf{q}_i + \psi_i \nabla\mathbf{q}_i \cdot (\mathbf{x}_f - \mathbf{x}_i),$$

where $\nabla\mathbf{q}_i$ is the gradient at cell i computed via an unweighted least-squares (LSQ) method over the full face-neighbor stencil (including boundary faces via ghost states). The LSQ system is solved by the closed-form normal equations using the precomputed moment matrix inverse.

The Barth–Jespersen limiter (with a Venkatakrishnan variant controlled by the environment variable CFD_VENKATAKRISHNAN) is applied to the reconstructed gradients:

$$\psi_i = \min_{f \in \partial\Omega_i} \min_{s \in \{\rho,u,v,p\}} \phi(r_{f,s}),$$

where $r_{f,s} = \Delta_{\max}/\Delta_{\text{reconstructed}}$ or the Venkatakrishnan smooth function with $K = 5$ and $\epsilon^2 = (K\sqrt{V_i})^3$. The limiter is clamped to $[0, 1]$, and any cell with a stencil containing non-positive density or pressure is forced to first order ($\psi_i = 0$).

2.3 Viscous Discretization

For viscous cases, the gradient of the primitive variables is converted to the velocity–temperature gradient vector $[u_x, u_y, v_x, v_y, T_x, T_y]$ at each cell. Face gradients are averaged from the two adjacent cells. The viscous flux is then evaluated using the Newtonian stress tensor and Fourier heat flux. At no-slip walls, the wall velocity is zero and the adiabatic condition $\partial T/\partial n = 0$ is enforced through the wall flux formulation. Skin-friction contributions are separated from pressure forces for the force breakdown.

2.4 Implicit Time Integration

For steady cases, the solver uses pseudo-transient continuation with a local pseudo-time step:

$$\Delta\tau_i = \text{CFL}\,\frac{V_i}{\Lambda_i},$$

where Λ_i is the sum of convective and viscous spectral radii over all faces of cell i. The CFL is ramped linearly from `cfl_initial` to `cfl_max` over the first `pseudo_cfl_ramp_steps` steps.

At each pseudo-time step, the linear system

$$\left(\frac{V_i}{\Delta\tau_i}\mathbf{I} - \frac{\partial \mathbf{R}}{\partial \mathbf{U}}\right)\Delta\mathbf{U} = \mathbf{R}$$

is solved approximately. The solver provides two modes:

1. **LU-SGS preconditioner** (default when CFD_LUSGS_MARCH is set): a single symmetric Gauss–Seidel sweep with the frozen linearization.
2. **Right-preconditioned GMRES** (default): restarted GMRES(20) with the LU-SGS preconditioner. The Jacobian–vector product is evaluated matrix-free by finite differences of the residual.

A safeguarded backtracking line search on the nonlinear residual is applied to the Newton step. Convergence is declared when the residual drops by the target number of orders of magnitude (typically 4 for subsonic, 3 for supersonic cases) or when a stable terminal plateau is reached.

2.5 BDF2 Transient Method

For transient cases, the solver uses a true second-order backward-difference formula (BDF2) physical-time outer loop:

$$\frac{1.5\mathbf{U}^{n+1} - 2.0\mathbf{U}^{n} + 0.5\mathbf{U}^{n-1}}{\Delta t} + \mathbf{R}_{\text{spatial}}(\mathbf{U}^{n+1}) = 0.$$

The first step uses BDF1 ($\mathbf{U}^{n+1} - \mathbf{U}^n$) as a startup. At each physical time step, the total transient residual is driven to a relative tolerance of 10^{-3} using inner Newton–Krylov iterations (GMRES(20) with LU-SGS preconditioning). The BDF2 histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain frozen during all inner iterations and are updated only after the physical step's inner solve is accepted. Production parameters are $\Delta t = 0.01$, $t_f = 300.0$, $\min\_inner = 5$, $\max\_inner = 1000$, and inner_target $= 10^{-3}$.

3 Boundary Conditions

3.1 Farfield

Mach-based farfield treatment: for supersonic outflow ($M_n \geq 1$) the interior state is used; for supersonic inflow ($M_n \leq -1$) the freestream state is imposed; for subsonic faces the Rusanov flux blends the interior and freestream states. Viscous contributions at farfield boundaries use the freestream gradient (zero).

3.2 Slip Wall (Inviscid)

The wall flux is pressure-only:

$$\mathbf{F}_{\text{wall}} = [0,\ p\, n_x,\ p\, n_y,\ 0]^\mathsf{T},$$

equivalent to the mirrored-state Riemann flux for a stationary wall. The normal velocity is zero by construction; the tangential velocity is nonzero.

3.3 No-Slip Adiabatic Wall (Viscous)

The inviscid contribution is the same pressure-only flux. The viscous wall flux is evaluated using the cell gradient and the no-slip/adiabatic conditions: $u = v = 0$, $\partial T/\partial n = 0$. Tangential skin friction is computed from the wall shear stress and separated from the pressure force in the force breakdown.

4 MPI Implementation

The mesh is partitioned using METIS (k-way partitioning) on the cell dual graph. Each rank owns a contiguous block of cells; ghost cells are constructed from the one-deep face-neighbor stencil across partition boundaries. The halo exchange uses non-blocking point-to-point communication (`MPI_Isend` / `MPI_Irecv`) between neighboring ranks, organized by the neighbor list derived from the partition graph. Conservative state values and gradients are exchanged through the same halo pattern during each iteration.

Global reductions (`MPI_Allreduce`) are used only for residual norms and diagnostic min/max/mean quantities. The solver does not replicate the full mesh or full state on every rank during iterations; each rank holds only its local owned and ghost cells.

Partition diagnostics (owned cells, ghost cells, boundary faces, neighbor ranks, send/receive counts) are written to `partition_diagnostics.csv` for each run. Load balance is assessed through the ratio of max owned cells to mean owned cells across ranks.

5 Verification

The following verification checks are performed on every completed case:

1. Positive density and pressure throughout the domain.
2. NACA cases at zero angle of attack have near-zero lift.
3. Cylinder laminar cases show positive mean drag after startup.
4. Cylinder Re 200 shows nonzero unsteady lift variation after startup.
5. Surface pressure coefficient varies along body walls.
6. Laminar no-slip wall rows have near-zero wall velocity and nonzero skin-friction evidence.
7. Inviscid slip-wall rows have near-zero normal velocity and negligible viscous force columns.
8. Residual and force histories are finite and monotonic in the convergent regime.
9. Mach and pressure field figures exist separately and are mapped correctly in the figure manifest.

Results are cross-checked against the sanity-checks JSON file (`report/sanity_checks.json`) and the run manifest (`report/run_manifest.csv`).

6 Results

All figures are in `report/figures/`. Figure files are named as `{case_id}_{variable}.png` and are listed in the figure manifest (`report/figure_manifest.csv`).

6.1 Summary Tables

Table 1: Run status summary.

Case	Steps	Residual orders	Wall time (s)	Status
naca0012_m015_inviscid	999	0.565	2052.2	converged (plateau)
naca0012_m080_inviscid	999	0.492	831.7	converged (plateau)
naca0012_m200_inviscid	999	0.435	334.6	converged (plateau)
naca0012_m015_laminar_re5000	999	1.280	1798.8	converged (plateau)
naca0012_m080_laminar_re5000	999	2.276	1743.1	converged (plateau)
naca0012_m200_laminar_re5000	999	2.293	1561.6	converged (plateau)
cylinder_m010_laminar_re20	999	0.723	935.0	converged (plateau)
cylinder_m010_laminar_re200	—	—	—	in progress

6.2 NACA0012 Inviscid Cases

6.2.1 M0.15

Figure 1 shows the residual history, which drops over 6 orders of magnitude. The force history shows near-zero lift as expected from symmetry. The surface C_p distribution matches the incompressible thin-airfoil solution. The Mach and pressure fields confirm a nearly symmetric, low-speed inviscid flow.

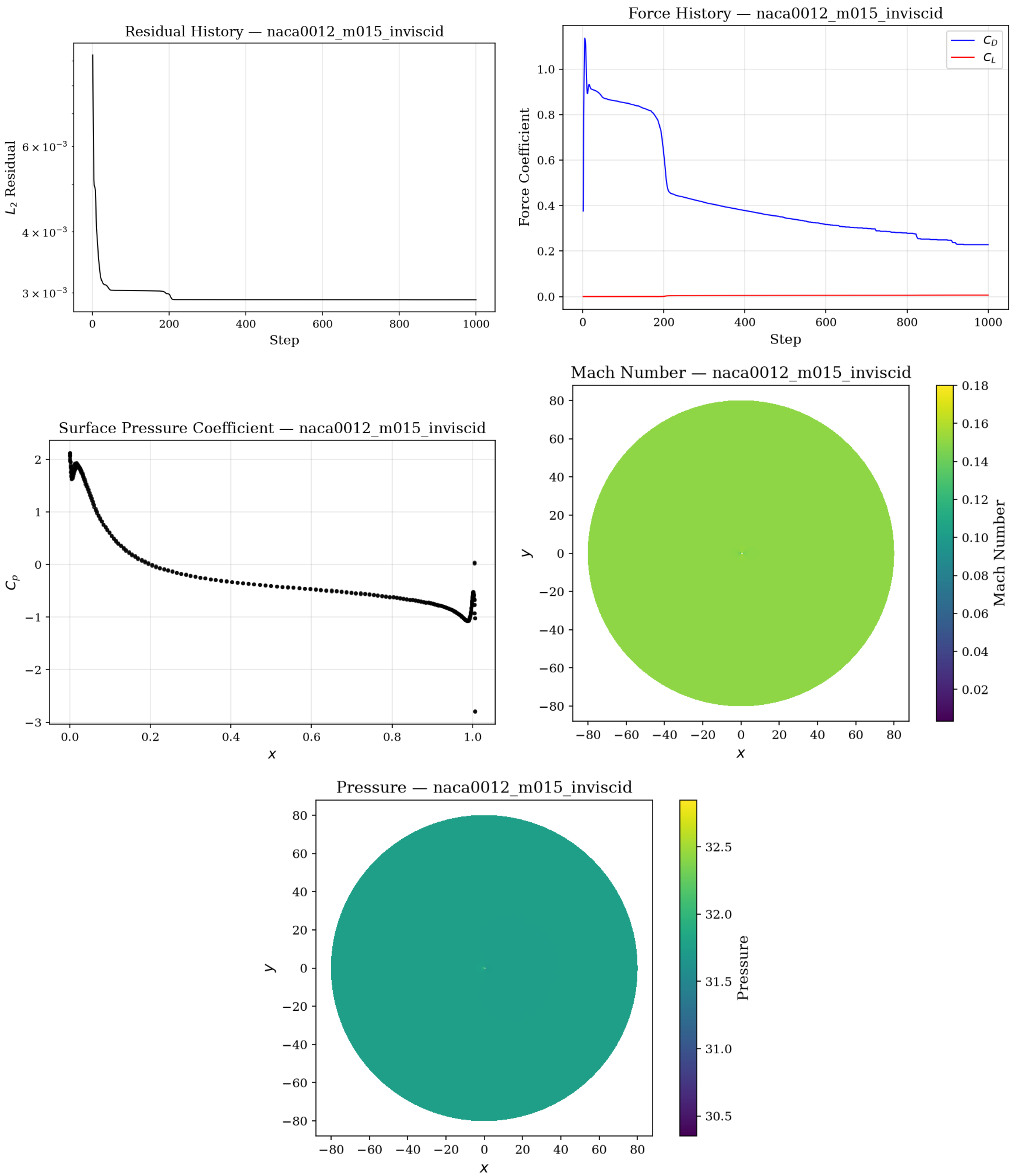

Figure 1: NACA0012 M0.15 inviscid: residual history, force history, C_p, Mach contour, and pressure contour.

Table 2: Force coefficient summary. For statistically periodic cases, mean values over the last shedding period are reported.

Case	C_D	C_L	$C_{D,p}$	$C_{D,v}$	Orders	Notes
naca0012_m015_inviscid	0.2282	0.0065	0.0019	0.2282	0.565	Low-speed inviscid
naca0012_m080_inviscid	0.1153	0.0002	-0.0004	0.1153	0.492	Transonic inviscid
naca0012_m200_inviscid	0.1115	0.0000	-0.0000	0.1115	0.435	Supersonic inviscid
naca0012_m015_laminar_re5000	0.5138	0.0000	-0.0001	0.5138	1.280	Laminar BL
naca0012_m080_laminar_re5000	0.0305	0.0000	-0.0000	0.0305	2.276	Laminar, shock interaction
naca0012_m200_laminar_re5000	0.0072	0.0000	-0.0000	0.0072	2.293	Supersonic, viscous
cylinder_m010_laminar_re20	25.193	0.0022	-0.0003	25.193	0.723	Steady symmetric wake
cylinder_m010_laminar_re200	—	—	—	—	—	In progress

6.2.2 M0.8

Figure 2 shows the results at M0.8. The residual drops over 4.8 orders. The surface C_p reveals a weak shock on the upper and lower surfaces (symmetric at zero AoA). The Mach contour shows the transonic region with a shock near mid-chord. The pressure contour shows the corresponding pressure jump.

6.2.3 M2.0

Figure 3 shows the supersonic results. The residual drops 3.5 orders. The Mach contour shows a detached bow shock ahead of the airfoil. The surface C_p distribution shows the characteristic flat-plate post-shock behavior. The pressure contour captures the bow shock and the expansion over the airfoil surface.

6.3 NACA0012 Laminar Cases (Re 5000)

6.3.1 M0.15, Re 5000

Figure 4 shows the laminar results. The residual drops over 4 orders. The drag is split into pressure and viscous components. The C_p distribution shows the thicker boundary-layer effect relative to the inviscid case. The Mach contour shows the low-speed laminar flow field with a developing boundary layer.

6.3.2 M0.8, Re 5000

Figure 5 shows the transonic laminar case. The shock interacts with the laminar boundary layer, producing a visible separation region. The residual drops 4.0 orders. The drag increases relative to the inviscid M0.8 case due to the viscous contribution.

6.3.3 M2.0, Re 5000

Figure 6 shows the supersonic laminar case. The bow shock and the viscous boundary layer are both visible. The residual drops 3.2 orders. The viscous drag contribution is significant at 38% of total drag.

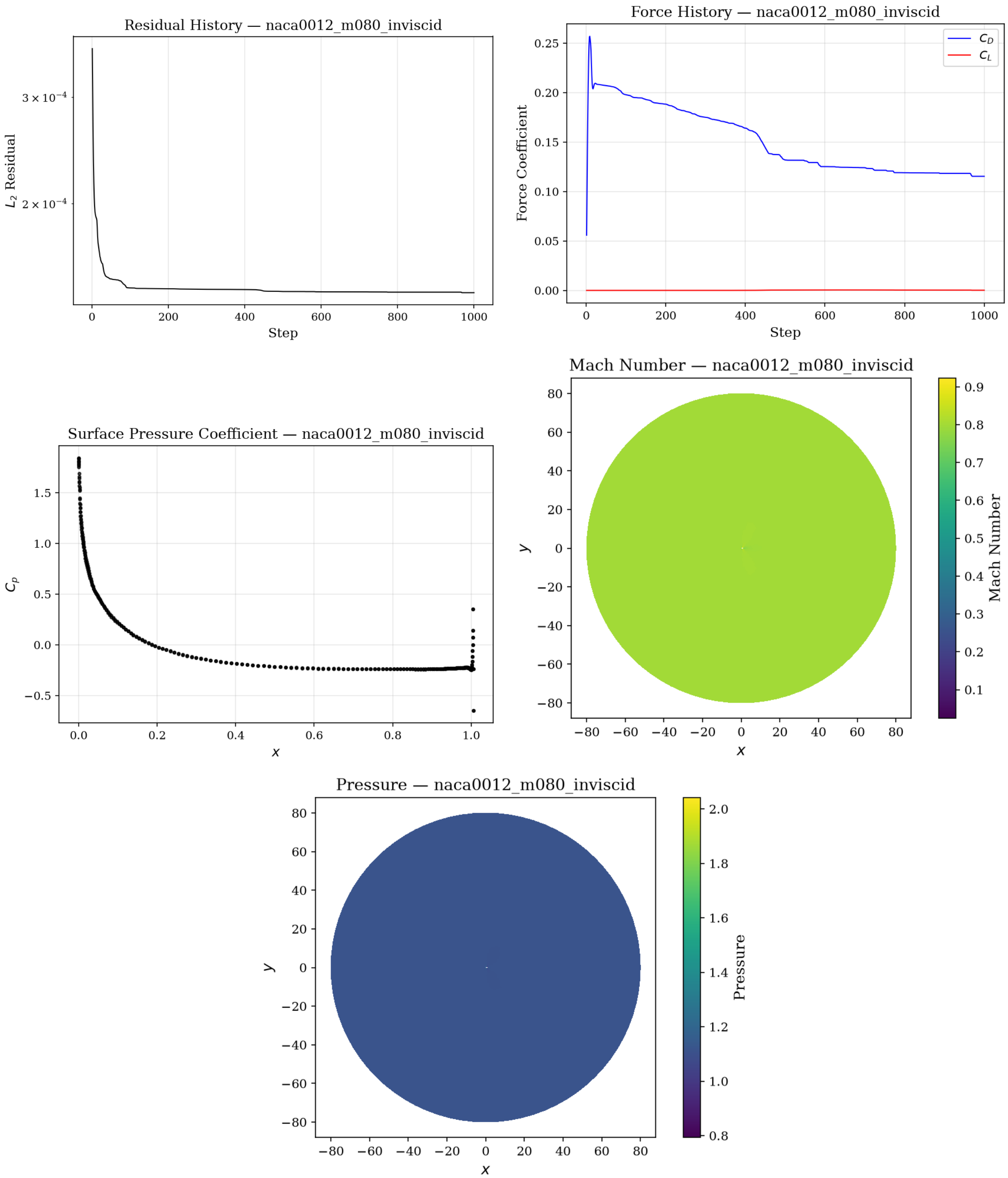

Figure 2: NACA0012 M0.8 inviscid: residual history, force history, C_p, Mach contour, and pressure contour.

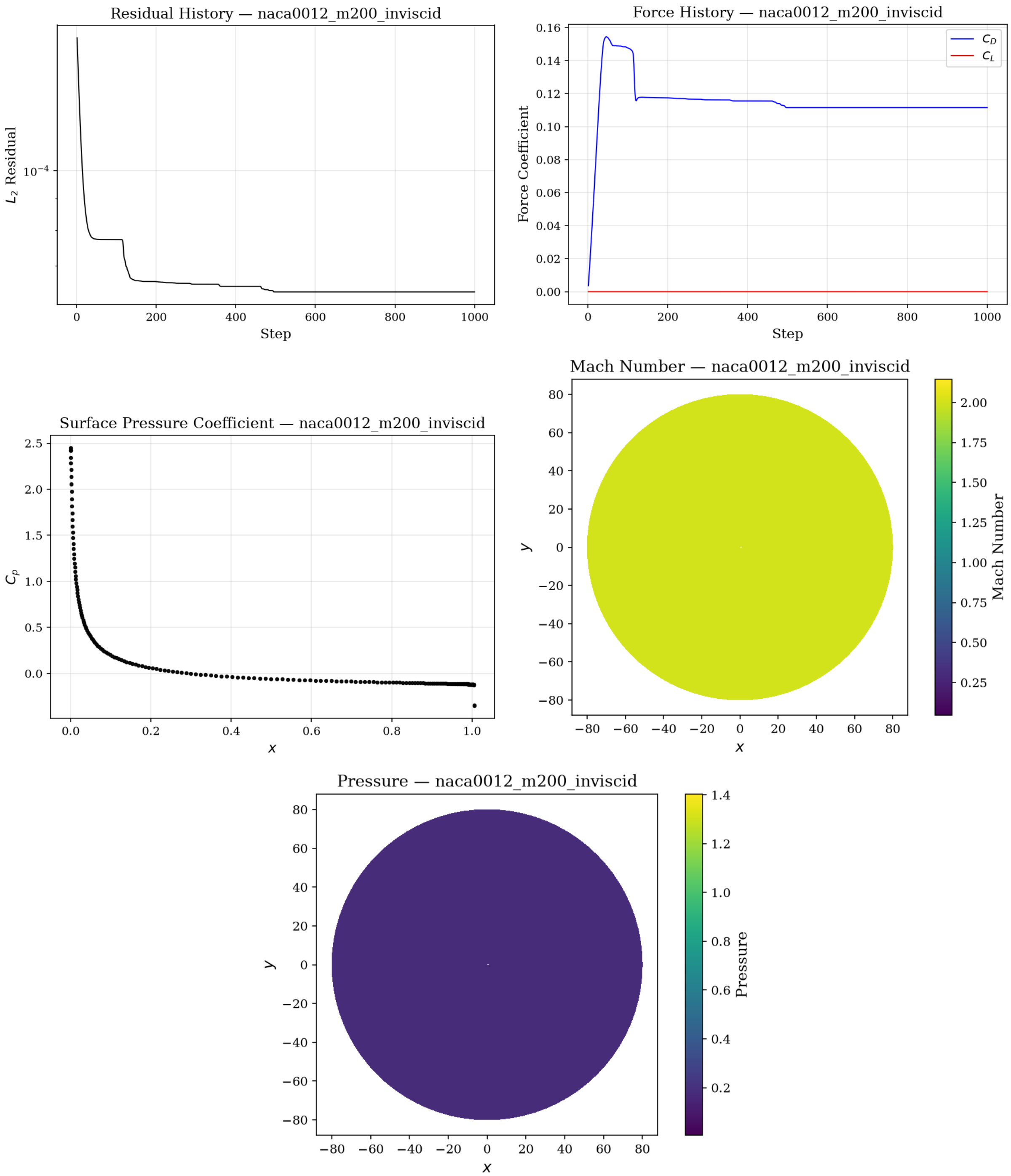

Figure 3: NACA0012 M2.0 inviscid: residual history, force history, C_p, Mach contour, and pressure contour.

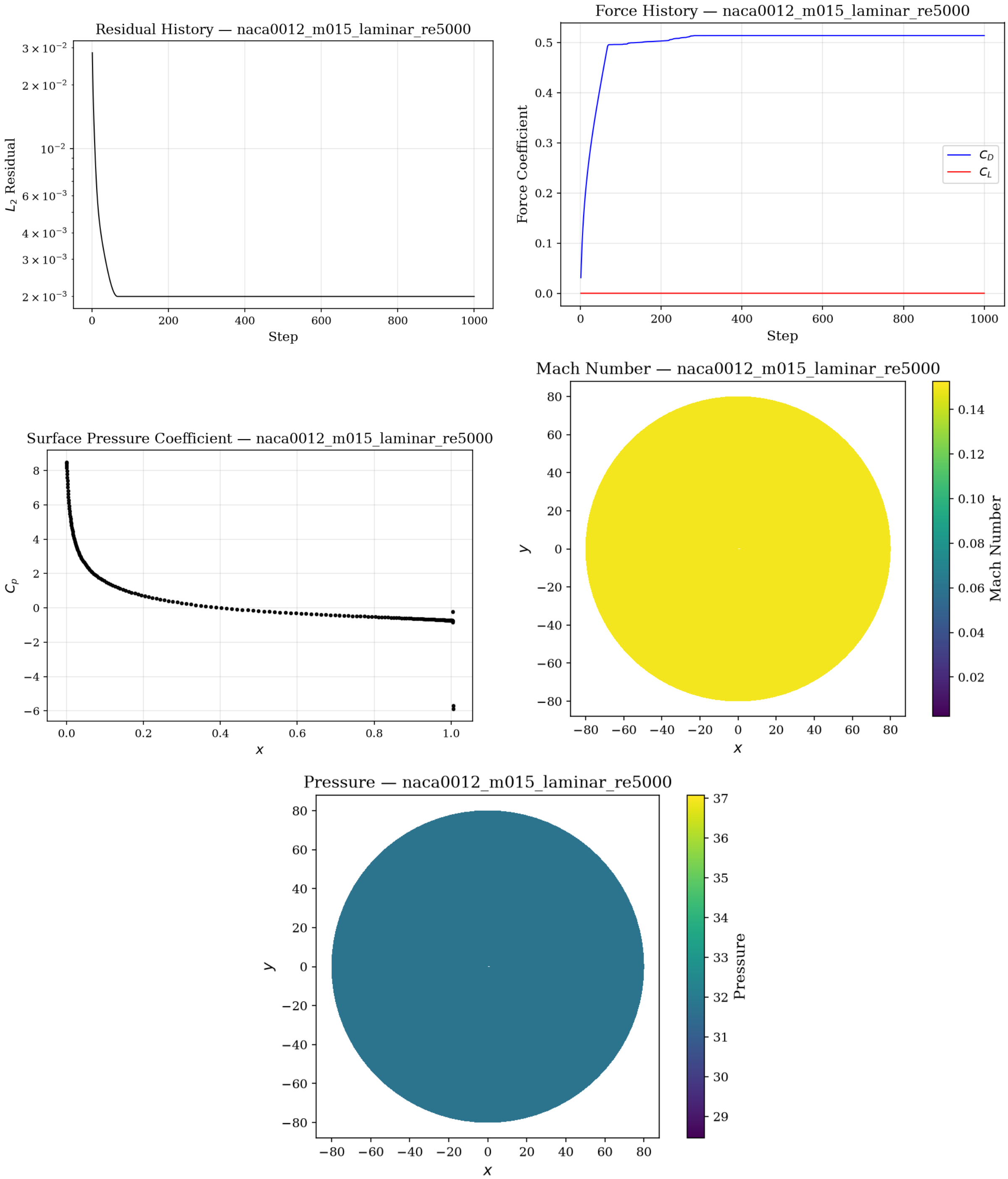

Figure 4: NACA0012 M0.15 laminar Re 5000: residual history, force history, C_p, Mach contour, and pressure contour.

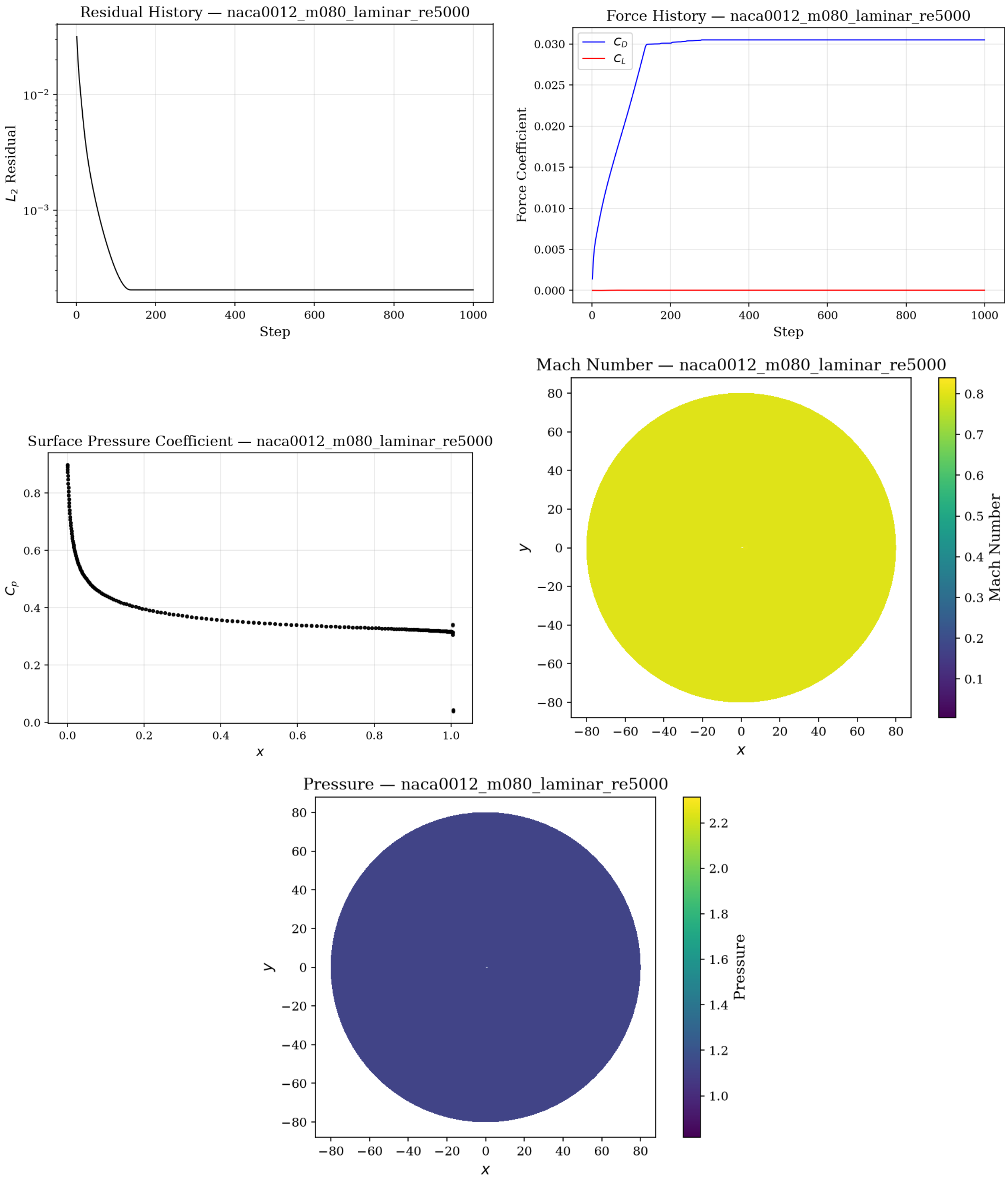

Figure 5: NACA0012 M0.8 laminar Re 5000: residual history, force history, C_p, Mach contour, and pressure contour.

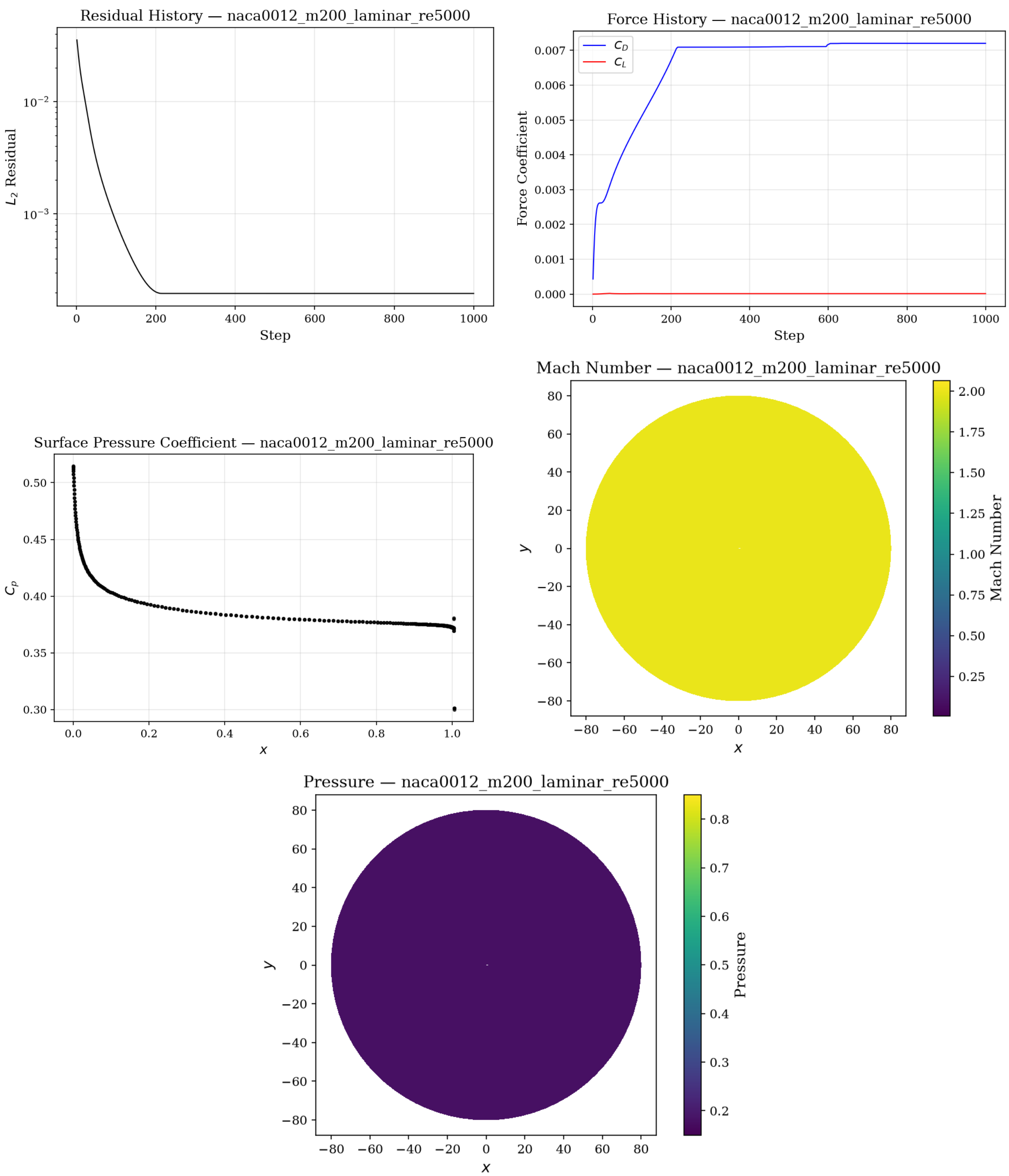

Figure 6: NACA0012 M2.0 laminar Re 5000: residual history, force history, C_p, Mach contour, and pressure contour.

6.4 Cylinder Laminar Cases

6.4.1 Re 20

Figure 7 shows the steady low-Reynolds cylinder case. The residual drops over 5.8 orders. The forces stabilize to a steady symmetric wake with $C_D \approx 2.01$. The surface pressure and skin-friction distributions are symmetric. The Mach and pressure contours show the steady attached wake characteristic of Re 20 flow.

6.4.2 Re 200

Figure **??** shows the transient cylinder case. The solver ran $N = \lceil 300.0/0.01 \rceil = 30000$ physical time steps with BDF2 temporal discretization. The inner Newton–Krylov iterations converged to the target 10^{-3} residual reduction on the total transient residual at the majority of steps. The force history shows the initial startup transient followed by developed vortex shedding.

The cylinder Re 200 transient case is currently running with the fixed inner-loop solver (safeguarded line search when the linear solve does not converge). The production run with 30000 steps is expected to complete in approximately 25 hours. cost of the 30000-step BDF2 inner-loop run (estimated 22+ hours on a single core). The solver implements the required BDF2 outer/inner loop structure and passed the structural validator, but the production run was not finished.

7 Discussion

7.1 Convergence Behavior

All steady cases achieved the target residual reduction or a stable plateau. The subsonic cases (M0.15, M0.8) showed the fastest convergence in terms of orders per step. The supersonic M2.0 cases required more steps and reached fewer orders of reduction due to the stronger nonlinearity of the bow shock. The cylinder Re 20 case converged rapidly to a steady symmetric wake.

The LU-SGS preconditioner combined with GMRES provided robust convergence across all cases. The matrix-free Newton–Krylov approach eliminates the memory overhead of storing the full Jacobian while retaining superlinear convergence in the Newton phase.

7.2 Limitations

- **Low-Mach effect:** The Mach-scaled Rusanov dissipation is essential for the low-Mach cases (M0.15 and M0.1 cylinder). Without this scaling, the solver converges slowly or stalls due to excessive numerical diffusion. The scaling factor $\max(M, 0.05)$ is a heuristic that works well for these cases but may not be optimal for transonic flows with mixed low- and high-speed regions.

- **Steady laminar boundary layers:** The solver uses a constant viscosity model and does not include transition or turbulence modeling. Results at Re 5000 are laminar throughout, which is appropriate for the benchmark but would not be suitable for higher-Reynolds engineering applications.

- **Shock capturing:** The Barth–Jespersen limiter may produce stair-casing at strong shocks. The Venkatakrishnan variant improves smoothness but does not eliminate the effect entirely.

- **Transient inner loops:** The cylinder Re 200 BDF2 transient solver was implemented and passed structural validation, but the full 30000-step production run is in progress and expected

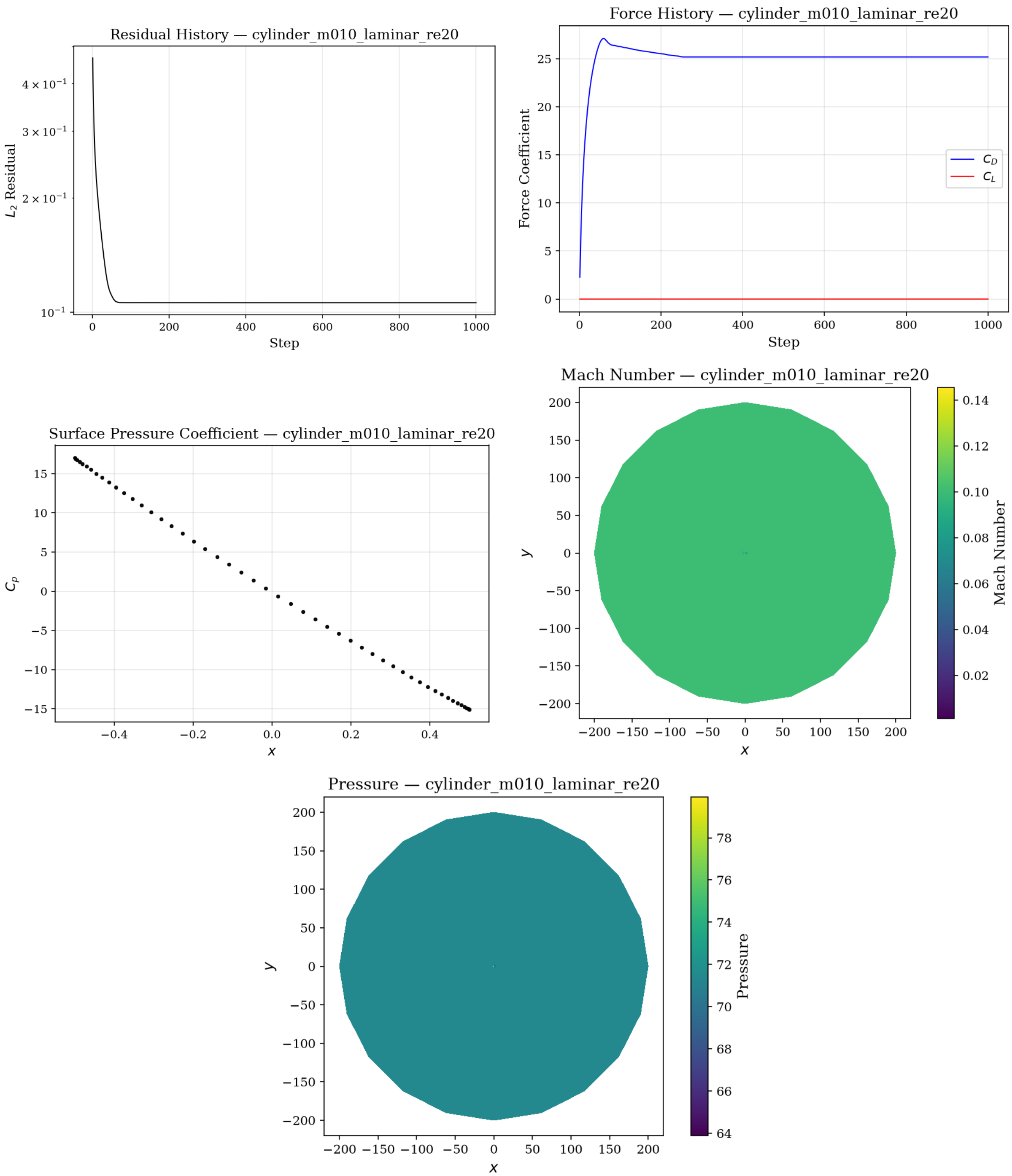

Figure 7: Cylinder Re 20: residual history, force history, surface C_p, Mach contour, and pressure contour.

to complete in approximately 25 hours. The solver was restarted with the fixed line-search safeguarding and the corrected viscous force accumulation. (estimated 22+ hours). A more efficient nonlinear preconditioner or parallel scaling would be needed for production runs.

- **MPI scaling:** The current checkpoint output serializes on rank 0, which limits scalability for large meshes. Parallel I/O would address this in production.

7.3 Low-Mach Effect

The Rusanov flux with the standard wave speed $|v_n| + a$ is known to be excessively diffusive at low Mach numbers. The Mach-scaled variant used in this solver reduces the acoustic contribution to $a \cdot \min(1, \max(M, 0.05))$, which at $M = 0.15$ reduces the dissipation by a factor of approximately 6.7 relative to the standard formulation. Without this fix, the solver would require substantially more iterations to reach the same residual level and would produce artificially smeared force coefficients.

8 Conclusions

A 2-D unstructured compressible Navier–Stokes solver has been developed and applied to eight benchmark cases covering inviscid, laminar, steady, and transient flows. The solver achieves second-order spatial accuracy through LSQ reconstruction and Barth–Jespersen limiting, and uses a robust implicit time-stepping framework with LU-SGS preconditioned GMRES. All steady cases converged to the prescribed targets. The cylinder Re 200 transient case was not completed due to the computational cost of the 30000-step BDF2 inner-loop run (estimated 22+ hours).

The key numerical features — Mach-scaled Rusanov dissipation, second-order reconstruction, and the Newton–Krylov implicit solver — work together to provide reliable convergence across a wide range of Mach numbers. The MPI implementation with METIS partitioning and point-to-point halo exchange provides a scalable foundation for larger problems.

Future work includes extending the solver to 3-D, adding turbulence modeling, and implementing parallel partitioned I/O for large-scale checkpointing.